

CS-PaperSum: A Large-Scale Dataset of AI-Generated Summaries for Scientific Papers

This paper presents a solution to a problem arising from the massive growth of scientific articles in computer science, which makes it difficult to track new research trends and extract key insights. To address this, **CS-PaperSum** is introduced as a large-scale dataset of over 91,000 articles drawn from 31 top-tier computer science conferences, enriched with structured summaries generated by AI using ChatGPT.

Throughout the article, it details the tools used to evaluate the quality of the summaries, verifies if the keywords were extracted correctly, and presents a case study on AI research trends.

Introduction

The introduction highlights that the rapid expansion of scientific literature in computer science has created significant challenges for researchers attempting to stay informed about new developments, track emerging trends, and synthesize insights from an overwhelming number of publications, making manual review impractical. It points out that open-source repositories like the Semantic Scholar Open Research Corpus (S2ORC) and the Microsoft Academic Graph (MAG) provide bibliographic information, abstracts, and citation networks, but lack structured summaries that capture the core contributions, methodologies, and future research directions.

Related Work

Several large-scale datasets (such as S2ORC and MAG) provide access to scientific articles and metadata, facilitating bibliometric studies, but they lack structured research summaries. The automatic summarization of scientific articles has been widely studied in NLP (Natural Language Processing), with recent advances leveraging neural abstractive summarization (such as **SciBERT** — which is a version of Google's BERT model pre-trained specifically on millions of scientific texts to accurately understand academic jargon — and **SPECTER** — Scientific Paper Embeddings using Citation-informed Transformers). CS-PAPERSUM extends this research by integrating summaries generated by ChatGPT, providing a new resource to evaluate LLM-based summarization in the scientific domain.

3.1 Data Collection and Content

Open-access articles were collected using the Semantic Scholar API, focusing on major computer science conferences. The final dataset encompasses 91,919 articles from 31 conferences (from 2017 to 2024), covering AI, computer vision, NLP, computational theory, and

architecture. For each article, the following was extracted: Title, Authors, Conference, Year, Abstract, and Citation Count. The conclusion extraction was performed via GROBID, converting PDFs into XML.

3.2 and 3.3 Statistics and Productive Affiliations

The distribution reveals high-impact conferences (NeurIPS, CVPR, ACL, etc.) and a steady increase in publications over the years. The affiliation analysis shows the dominance of elite universities (MIT, Stanford, CMU, etc.) and a strong presence of industry labs (Google, DeepMind, Microsoft).

3.4 Influential Conferences

Influence, measured by the median number of citations, places AI and computer vision conferences (CVPR, ICLR, ICCV) at the top. Beyond AI, security conferences (NDSS, USENIX) and programming languages/architecture (PLDI, ASPLOS) demonstrate substantial influence, underlining the importance of cybersecurity and parallel computing.

4. Summarizing the Papers

To enhance utility, structured summaries were generated for each article using ChatGPT (GPT-3.5). The model was prompted to extract the main insights in a structured list format based on the title, abstract, and conclusion. The standard format includes: Key Takeaways, Importance, Proposed Model/Method, Performance, Effectiveness, and Future Work.

4.1 Quality Assessment

Reliability was evaluated via embedding alignment (using SciBERT and t-SNE) and keyword overlap analysis (using KeyBERT). The results indicate that the summaries strongly retain the original semantics, characteristics, and keywords, validating ChatGPT's effectiveness in this domain.

4.2 Case Study: Identifying Research Trends

Trends were analyzed in NeurIPS, AAAI, EMNLP, and ICCV. In NeurIPS, the focus migrated from classic reinforcement learning to Graph Neural Networks (GNNs) and self-supervised learning. In AAAI, there is a growing focus on trustworthy AI, ethics, and human-AI collaboration. EMNLP drastically shifted from statistical translation to transformers, RAG (Retrieval-Augmented Generation), and instruction tuning. In ICCV, the field transitioned from classic object detection to neural radiance fields (NeRFs) and diffusion models.

5. Conclusion

CS-PAPERSUM was introduced as an enriched dataset that provides standardized summaries to bridge the gap between bibliographic data and actionable insights. The dataset enables trend analysis and provides a foundation for future systems based on RAG and multi-document summarization for scientific discovery.